

In-Class Worksheet - ANSWER KEY

Introduction to R and Data Presentation

Course: H524 Introduction to Biostatistics

Topic: Introduction to R, Data Presentation, Numerical Summary Measures

1 Part 1: Getting Started with R

1.1 Exercise 1.1: Basic R Operations

Work with a partner to complete these basic R calculations:

```
1 # 1. Use R as a calculator
2 3 * 2 + 4
3 # Answer: 10
4
5 # 2. Create a simple vector of systolic blood pressure readings
6 blood_pressure <- c(170, 193, 110, 135, 135, 115)
7 # This creates a numeric vector with 6 blood pressure values
```

Discussion Questions - Answers:

- **How do you create a vector in R?** Use the `c()` function with values separated by commas
- **What does the `c()` function do?** It combines (concatenates) values into a vector

2 Part 2: Data Types and Classification

2.1 Exercise 2.1: Identifying Data Types

For each variable below, classify it as:

- **Nominal** (categorical without order)
- **Ordinal** (categorical with order)
- **Discrete** (countable numbers)
- **Continuous** (measurable numbers)

Variable	Type	Reasoning
Blood type (A, B, AB, O)	Nominal	Categories with no natural order
Pain level (none, mild, moderate, severe)	Ordinal	Categories with clear ordering from least to most severe
Number of hospitalizations this year	Discrete	Countable whole numbers (0, 1, 2, 3, ...)
Body temperature (°F)	Continuous	Can be measured to any level of precision
Treatment group (control, low dose, high dose)	Nominal	Categories with no inherent order

2.2 Exercise 2.2: AI Enhancement Activity

Using AI Tools: Use GitHub Copilot or another AI assistant to help you:

1. Generate R code to create vectors for each data type above
2. Ask AI: “What’s the best way to store categorical data in R?”

Sample AI-Generated Code:

```
1 # Nominal data - blood type
2 blood_type <- factor(c("A", "B", "AB", "O"))
3
4 # Ordinal data - pain level
5 pain_level <- factor(c("none", "mild", "moderate", "severe"),
6                       levels = c("none", "mild", "moderate", "severe"),
7                       ordered = TRUE)
8
9 # Discrete data - hospitalizations
10 hospitalizations <- c(0, 1, 2, 1, 0, 3)
11
12 # Continuous data - temperature
13 temperature <- c(98.6, 99.2, 97.8, 100.1, 98.9)
14
15 # Nominal data - treatment group
16 treatment <- factor(c("control", "low_dose", "high_dose"))
```

AI Answer: Use `factor()` for categorical data, with `ordered = TRUE` for ordinal data.

3 Part 3: Descriptive Statistics

3.1 Exercise 3.1: Measures of Central Tendency

Using the blood pressure data: `blood_pressure <- c(170, 193, 110, 135, 135, 115)`

Complete the calculations both by hand and in R:

```
1 # Create the data
2 blood_pressure <- c(170, 193, 110, 135, 135, 115)
3
4 # Calculate mean
5 mean_bp <- mean(blood_pressure)
6 mean_bp # 143
7
8 # Calculate median
9 median_bp <- median(blood_pressure)
10 median_bp # 135
11
12 # Find the mode (most frequent value)
13 table(blood_pressure) # Shows 135 appears twice
14 # Mode: 135
15
16 # Order the data
17 ordered_bp <- sort(blood_pressure)
18 ordered_bp # 110 115 135 135 170 193
```

Hand Calculations:

- Mean: **143** mmHg $(170+193+110+135+135+115)/6 = 858/6 = 143$
- Median: **135** mmHg $(135+135)/2 = 135$ (middle values)
- Mode: **135** mmHg (appears twice)

R Results:

- Mean: **143** mmHg
- Median: **135** mmHg
- Mode: **135** mmHg

4 Part 4: Measures of Variability

4.1 Exercise 4.1: Spread of Data

```
1 # Create the data
2 blood_pressure <- c(170, 193, 110, 135, 135, 115)
3
4 # Calculate range
5 max_bp <- max(blood_pressure) # 193
6 min_bp <- min(blood_pressure) # 110
7 range_bp <- max_bp - min_bp   # 83
8
9 # Calculate variance
10 var_bp <- var(blood_pressure) # 1116.4
11
12 # Calculate standard deviation
13 sd_bp <- sd(blood_pressure)   # 33.41
14
15 # Calculate standard error of the mean
16 n <- length(blood_pressure)   # 6
17 sem_bp <- sd_bp / sqrt(n)     # 13.64
```

Fill in your results:

- Range: **83** mmHg (193 - 110)
- Variance: **1116.4** mmHg² (sample variance)
- Standard Deviation: **33.41** mmHg (square root of variance)
- Standard Error of Mean: **13.64** mmHg ($33.41/\sqrt{6}$)

Discussion: These measures tell us the blood pressure data has considerable spread (range = 83), high variability (SD = 33.41), and the mean has substantial uncertainty (SEM = 13.64).

5 Part 5: AI-Enhanced Data Exploration

5.1 Exercise 5.1: Using AI for Data Summary

Prompt for AI: “Help me create a comprehensive summary of this blood pressure dataset: c(170, 193, 110, 135, 135, 115). Include all relevant descriptive statistics and suggest appropriate visualizations.”

Sample AI-Generated Code:

```
1 # Blood pressure data analysis
2 blood_pressure <- c(170, 193, 110, 135, 135, 115)
3
4 # Comprehensive summary
5 summary(blood_pressure)
6 #   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
7 #   110     125     135     143     158     193
8
9 # Additional statistics
10 length(blood_pressure)    # n = 6
11 sd(blood_pressure)        # SD = 33.41
12 var(blood_pressure)       # Var = 1116.4
13 range(blood_pressure)     # Range: 110 193
14 IQR(blood_pressure)       # IQR = 33
15
16 # Suggested visualizations
17 hist(blood_pressure, main="Blood_Pressure_Distribution",
18      xlab="Systolic_BP_(mmHg)", col="lightblue")
19
20 boxplot(blood_pressure, main="Blood_Pressure_Boxplot",
21         ylab="Systolic_BP_(mmHg)")
22
23 dotchart(blood_pressure, main="Blood_Pressure_Dot_Plot")
```

Verification Steps - Sample Answers:

- **Run the code and check if results match:** Yes, all calculations match hand calculations
- **Do the suggested visualizations make sense?** Yes, histogram shows distribution, boxplot shows quartiles and outliers, dot plot shows individual values
- **Are there any errors or improvements needed?** Code is correct, might add axis labels for clarity

What you learned from AI - Sample Answers:

- New R functions discovered: **summary()**, **IQR()**, **dotchart()**
- Visualization suggestions: **histogram**, **boxplot**, **dot chart**
- Any surprising insights: **IQR = 33**, shows middle 50% spread

6 Wrap-up Discussion

Reflection Questions - Sample Answers:

1. **When might you prefer mean vs. median?** Use median when data is skewed or has outliers; use mean when data is normally distributed

2. **How can AI tools help while ensuring accuracy?** AI generates code quickly, but always verify results by hand or with known datasets; check for logical errors
3. **What's one new R function you learned?** Examples: `factor()`, `summary()`, `IQR()`, `table()`

Next Class Preview: We'll explore data visualization with histograms and boxplots, plus dive deeper into R programming basics.

7 AI Usage Documentation - Sample

Sample documentation:

- AI tools used: **GitHub Copilot, ChatGPT**
- Key prompts: **"Generate R code for data summary", "Best way to store categorical data"**
- Verification methods: **Hand calculations, checking output against expected results**
- Errors found and corrected: **None in this exercise, but always double-check factor levels**